IOT IMPLEMENTATION OFTHE SEMESTRAL PROJECT

PROJECT REQUIREMENTS

SYSTEM INTEGRATIONAND ARCHITECTURE

AUTOMATED RFID/BIOMETRIC ATTENDANCE MANAGEMENT SYSTEM FOR IT STUDENTS

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Statement of the Problem

The problem we want to address is the difficulty of managing attendance manually. It takes time, can lead to errors, and makes it difficult to access and analyze attendance records in real time. Our system solves this by using RFID and biometric verification, cloud-synced records, and attendance analytics.

The Problem it Solves

The system helps solve problems with traditional student attendance. Manual attendance takes a lot of time and can lead to mistakes or missing records. It can also be difficult to make sure that the correct student is marked present. Attendance records may not be available right away, making it harder for teachers to monitor students. It can also be difficult to review and understand attendance patterns. Our system solves these problems by automatically recording attendance, verifying students using RFID and biometrics, saving records to the cloud, and providing attendance reports and analytics.

Hardware Components

Microcontroller – Arduino Uno R3

Biometric Sensor – R307/AS608

Timekeeping – DS3231 RTC

Display Unit – 16 X LCD + I2C

Data Storage – MicroSD Card Module

Audio/Visual Alert – Buzzer & LEDs

The Arduino Uno R3 serves as the main brain of the system because it controls and connects all the other components. The R307/AS608 biometric sensor is used to scan and identify a person’s fingerprint. The DS3231 RTC keeps the correct date and time, which is important for recording attendance. The 16×2 LCD with I2C displays information such as the user’s name, time, or attendance status. The MicroSD Card Module is used to save attendance records and other important data. Lastly, the buzzer and LEDs provide audio and visual alerts to indicate whether the fingerprint scan was successful or if there was an error.

Sensor Integration

The proposed system uses the R307/AS608 Biometric Sensor as its main sensor to scan and identify a person’s fingerprint. When the fingerprint is recognized, the Arduino Uno R3 processes the information and records the date and time using the DS3231 RTC. The attendance record is then saved to the MicroSD Card, while the 16×2 LCD shows the result. The buzzer and LEDs give a sound and light signal to let the user know if the fingerprint was successfully scanned.

Device Programming

The Arduino controls the biometric attendance system. It receives the fingerprint from the R307/AS608 sensor and checks if it matches a registered user. If the fingerprint is recognized, the system gets the current date and time from the DS3231 RTC. The attendance result is shown on the LCD, while the green LED and buzzer indicate a successful scan and the red LED indicates an unrecognized fingerprint. The attendance record is also saved on the MicroSD card. For the IoT part, an ESP8266 or ESP32 connects the system to Wi-Fi and sends the attendance data to the server using HTTP REST API or MQTT. If the Wi-Fi connection is unavailable, the attendance record is saved on the MicroSD card first and can be sent to the server when the connection is available again.

Backend Integration

The backend application receives the attendance data from the IoT device through the HTTP REST API. The data includes the student's ID, name, date, time, and attendance status. After receiving the data, the backend saves the information in the database. The saved attendance records can then be viewed on the web dashboard by authorized users.

System Architecture